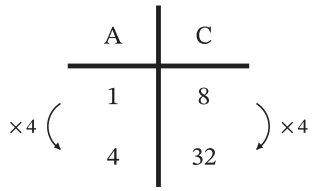
**11** There are 30 children in a nursery school.
At least 1 adult is needed for every 8 children in the nursery.

(a) Work out the least number of adults needed in the nursery.




A = adults; C = children.

3 adults cover only 24 children.

**1 mark**
Compare multiples of 8

**1 mark**
Need 4 adults

4

**(2)**

2 more children join the nursery.

(b) Does this mean that more adults are needed in the nursery?
You must give a reason for your answer.

No. There are now 32 children.

4 adults can supervise 4 × 8 = 32 children.

**1 mark**
The existing 4 adults are sufficient

**(1)**

**(Total for Question 11 is 3 marks)**

**12** Emma has 45 rabbits.

30 of the rabbits are male.
8 of the female rabbits have short hair.
12 of the rabbits with long hair are male.

(a) Use the information to complete the two-way table.

| | Male | Female | Total |
|---|---|---|---|
| **Long hair** | | | |
| **Short hair** | | | |
| **Total** | | | |

**(3)**

One of Emma's rabbits is chosen at random.

(b) Write down the probability that this rabbit is a female with short hair.

**(1)**

**(Total for Question 12 is 4 marks)**

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